Data Visualization & Analytics
Final Project Report

LinkedIn Job Market Demand, Salary & Competition Intelligence

Project Title	LinkedIn Job Market Demand, Salary & Competition Intelligence
Sector	Human Resources Analytics / Recruitment Intelligence / Labour Market Analytics
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Institute	Newton School of Technology
GitHub Repository URL	https://github.com/PALAK7890/SectionA_Team14_JobMarketTrends
Tableau Public URLs	Job Demand Intelligence: https://public.tableau.com/app/profile/palak.5572/viz/JobMarketDemandIntelligence/Dashboard1?publish=yes Salary Intelligence: https://public.tableau.com/app/profile/sarvesh.kumar6839/viz/Dashboard2_17773708580510/SalaryCompensationIntelligence?publish=yes Competition Intelligence: https://public.tableau.com/views/Krishnasdashboard/Dashboard4
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2. Executive Summary

This project studies the LinkedIn job market from a decision-maker perspective: where hiring demand is concentrated, which roles attract candidates, how salary transparency varies, and which job categories face stronger competition or hiring difficulty. The analysis uses a Python-based ETL and statistical workflow, followed by Tableau dashboards designed for executive review and operational drill-down.

The business challenge is that recruiters, job platforms, and workforce planners cannot rely only on job counts. A category may have many postings but weak candidate attention, while a smaller category may show high salary and high engagement. This report therefore combines demand, salary, competition, efficiency, geography, and experience-level views.

- 120,484 postings show average engagement of 14.63 views and 10.54 applications per posting, with an average conversion rate of 17.52%.
- Only 29.12% of postings include usable salary information, which makes salary transparency a major market gap.
- Most jobs are low-demand: 117,827 postings are in the low-demand segment, compared with 2,390 medium and 235 high-demand postings.
- Data & Analytics and Technology show the strongest salary signals, while Marketing and HR & Recruiting lead candidate engagement and efficiency.
- Remote jobs show a salary and competition premium, requiring separate hiring strategy.

Highest-impact recommendations are to improve salary disclosure, optimize low-demand job listings, benchmark pay for technical roles, and create separate remote hiring funnels.

3. Sector & Business Context

The project belongs to HR analytics and labour market intelligence. Modern hiring teams operate in a noisy market where raw job count does not reveal whether roles are attractive, competitive, efficiently filled, or fairly compensated. The primary decision-makers are recruitment leaders, HR business partners, workforce planners, job-platform strategy teams, and students evaluating market opportunities.

The business value is better hiring resource allocation, improved compensation strategy, stronger candidate targeting, and higher job-listing efficiency. A dashboard-driven system lets stakeholders move from broad market observation to specific category, state, salary, and remote-work decisions.

4. Problem Statement & Objectives

Formal problem definition: Analyze LinkedIn job posting data and build an executive-ready analytics solution that explains job demand, candidate engagement, salary patterns, remote-work effects, and hiring competition across categories, locations, and experience levels.

1. Clean and transform raw LinkedIn job posting data into a reliable Tableau-ready dataset using Python.
2. Measure demand using job counts, views, applications, conversion rate, and demand segments.
3. Analyze salary by experience level, category, remote label, and salary bracket.
4. Evaluate competition and hiring efficiency using competition score, days active, application intensity, and hard-to-fill indicators.
5. Design Tableau dashboards for executive review and operational filtering.

Scope includes LinkedIn postings, company attributes, salary fields, applications, views, geography, work type, experience level, and job category. It does not measure confirmed hiring outcomes or candidate demographics.

5. Data Description

Dataset source: Kaggle LinkedIn Job Postings dataset by arshkon/linkedin-job-postings, accessed through kagglehub in Python. The final analytical version is referenced in the GitHub repository as postings_tableau_ready.csv and related Tableau-ready outputs.

Source / field	Role in analysis
postings.csv	Main job-level table containing job ID, title, location, work type, salary, views, applications, and dates.
companies.csv	Adds company names and company size.
employee_counts.csv	Adds latest employee count as a scale indicator.
company_industries.csv + mapping	Maps companies to readable industry labels.
job_skills.csv + mapping	Links jobs to skill labels for future skill-demand analysis.
Final dashboard dataset	120,484 job-level records and 26 analytical columns in the dashboard-ready version.

Key fields include job_id, company_name, title, job_category, formatted_experience_level, formatted_work_type, remote_label, location, state, normalized_salary, salary_bracket, pay_period, views, applies, conversion_rate, days_active, application_type, listing_month, listing_year, and ordering fields for Tableau.

- LinkedIn may overrepresent digital, corporate, and urban roles.
- Salary is missing for a large share of records; salary analysis is based only on postings with usable salary data.
- Views and applications indicate engagement, not final hiring.
- Listing-month distribution is concentrated around April, so time trends should be interpreted carefully.

6. Data Cleaning & ETL Pipeline

All primary cleaning and transformation steps were executed in Python. The pipeline followed ingestion, merging, cleaning, feature engineering, and export.

Kaggle Data → Cleaning → Feature Engineering → EDA → Statistical Analysis → Tableau Dashboards

Stage	Method used	Purpose
Ingestion	Downloaded Kaggle dataset using kagglehub and loaded source CSVs with pandas.	Built the raw data foundation.
Merging	Joined postings with company metadata, latest employee counts, industry labels, and skills.	Created enriched job-level data.
Duplicate/outlier handling	Dropped duplicates and filtered normalized_salary to the 5th-95th percentile while retaining missing salary rows.	Reduced noise and extreme values.
Type conversion	Converted numeric fields and parsed time fields from milliseconds to datetime.	Prepared data for analysis and Tableau.

Text standardization	Trimmed text fields and standardized missing-like values.	Improved dashboard consistency.
Feature engineering	Created year, month, is_remote, demand_score, experience_group, final_company_name, city, state, location_remote_flag, and log_salary.	Enabled demand, remote, location, and salary analysis.
Export	Saved enriched job-level and exploded skills CSVs.	Created Tableau-ready files.

Views and applications were filled where appropriate for demand-score calculations. Salary was not blindly imputed because missing salary itself is a transparency signal. Missing salary is therefore measured through Salary Info %, which is 29.12%.

7. KPI & Metric Framework

KPI	Computation logic	Business meaning
Total Job Posting	Count of job_id	Market supply and coverage.
Average Views	Mean views per posting	Listing visibility.
Average Applies	Mean applications per posting	Candidate action.
Conversion Rate	Applications divided by views	How effectively views convert into applications.
Average Salary	Mean normalized_salary for disclosed salary rows	Compensation benchmark.
Salary Info %	Share with non-null normalized_salary	Salary transparency.
Competition Score	applies / views	Applicant competition intensity.
Efficiency Score	applies / days_active	Application speed.
Hard-to-fill %	Weak response relative to active days	Hiring friction indicator.
Demand Segment	Low/medium/high demand grouping	Prioritization of opportunity.

8. Exploratory Data Analysis (EDA)

EDA was performed across demand, salary, category, location, time, remote status, and experience segments. Demand and engagement analysis showed moderate average engagement, and the views-vs-applications scatter confirms a positive relationship between visibility and candidate action. Some high-view jobs still receive fewer applications, indicating possible mismatches in job relevance, salary, role clarity, or employer attractiveness.

Category analysis revealed that posting volume is concentrated in broad categories such as Other, Management, Technology, Healthcare, Sales, and Finance & Accounting. Candidate engagement is not identical to volume: Marketing, HR & Recruiting, Data & Analytics, Technology, and Finance & Accounting show stronger view intensity.

Geographic analysis showed stronger demand in major economic states such as California, Texas, and Florida. Salary analysis showed a clear experience premium, with Executive and Director roles at the top and Entry-level/Internship at the bottom. Remote-work analysis indicates remote jobs offer a compensation premium but also higher competition intensity.

9. Statistical Analysis

The statistical notebook tested whether salary differences across remote status, category, and experience level were meaningful rather than random dashboard noise.

Analysis	Result	Interpretation
Remote vs on-site salary t-test	p-value = 2.6097e-196	Remote and on-site salary distributions differ significantly.
Salary by job category ANOVA	p-value = 0.0	Average salary differs significantly across categories.
Category salary means	Data & Analytics approx. 144,383; Technology approx. 131,120; Management approx. 118,108; Marketing approx. 102,936.	Analytical and technical categories occupy stronger pay positions.
Efficiency distribution	Mean = 0.3223 applications per active day; median = 0.069.	Most roles accumulate applications slowly, while a few receive fast response.
Demand segmentation	Low = 117,827; Medium = 2,390; High = 235.	High-demand jobs are rare and should be studied as best-practice examples.

10. Tableau Dashboard Design

The Tableau solution is organized into three decision layers: Job Market Demand Intelligence, Salary & Compensation Intelligence, and Competition & Opportunity Intelligence. Together they support executive review, recruiter planning, and candidate market interpretation.

Dashboard 1 objective: identify where job demand, views, applications, conversion, geography, and demand segments are strongest.

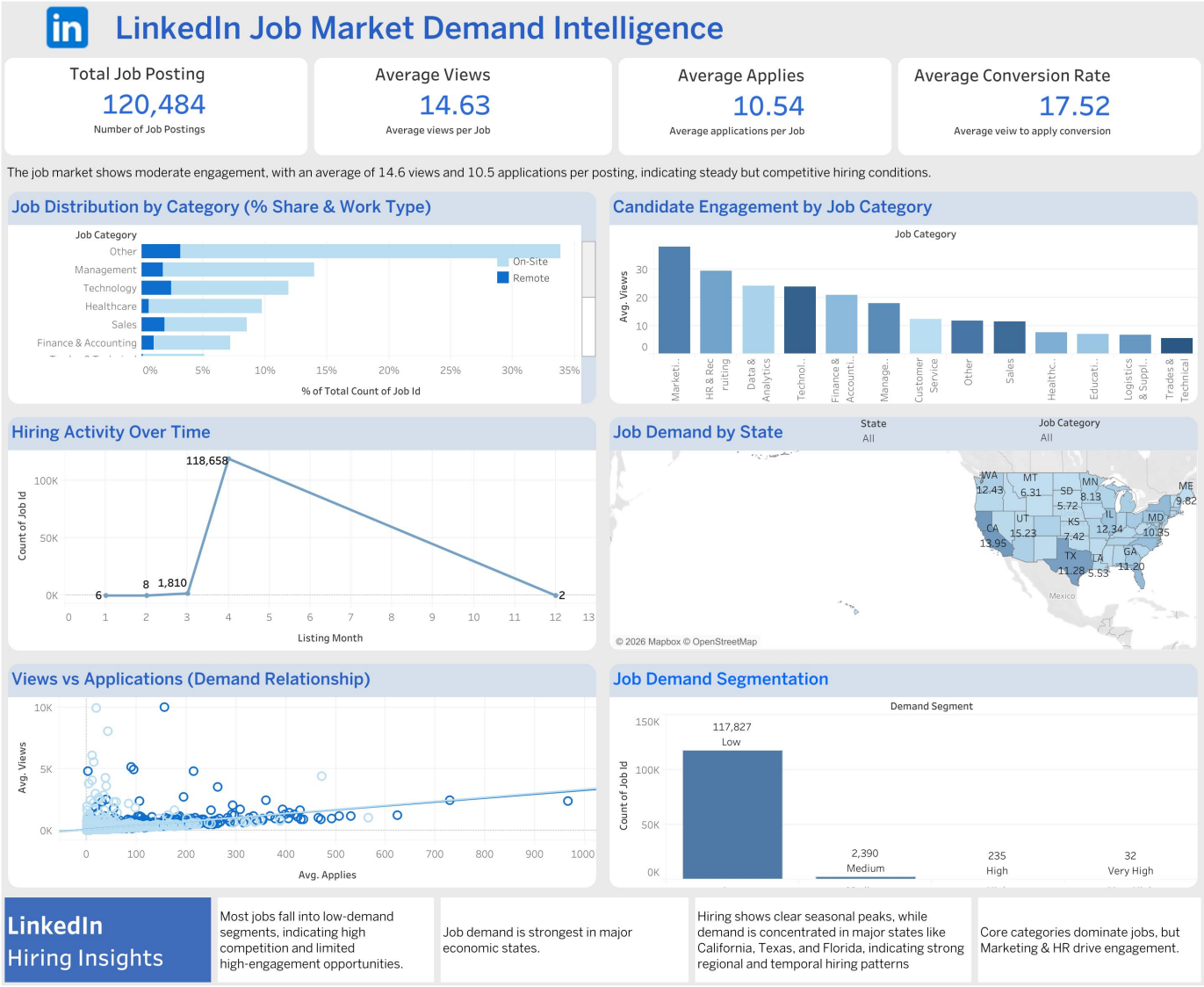

Figure 1. Job Market Demand Intelligence - KPI cards, category distribution, engagement, time trend, and state map with demand relationship, segmentation, and insight tiles.

Dashboard 2 objective: understand salary levels, salary distribution, experience premium, remote salary patterns, and salary bracket concentration.

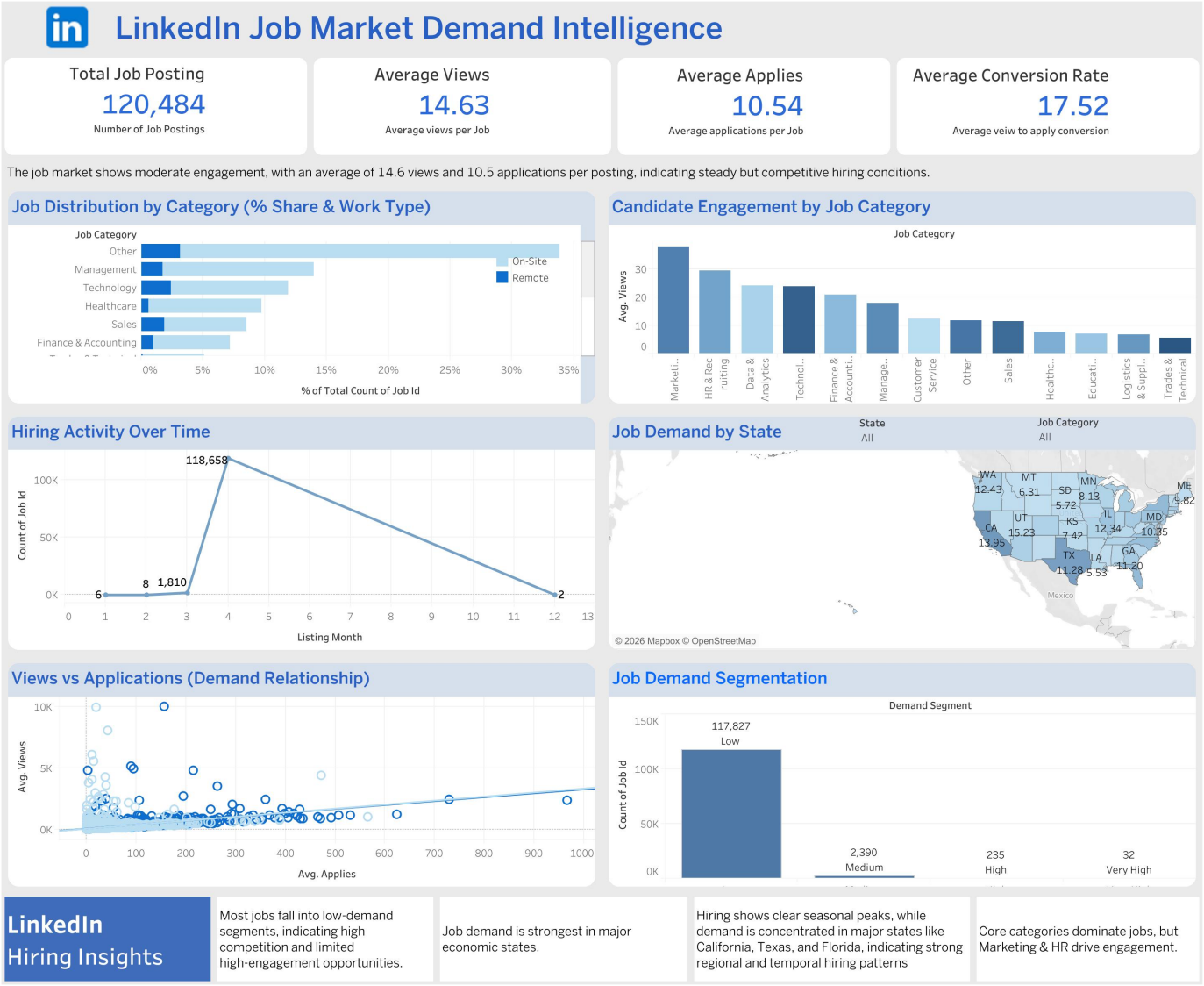

Figure 2. Salary & Compensation Intelligence dashboard.

Dashboard 3 objective: compare competition intensity, application efficiency, remote-vs-onsite competition, demand-salary positioning, and hiring difficulty by category.

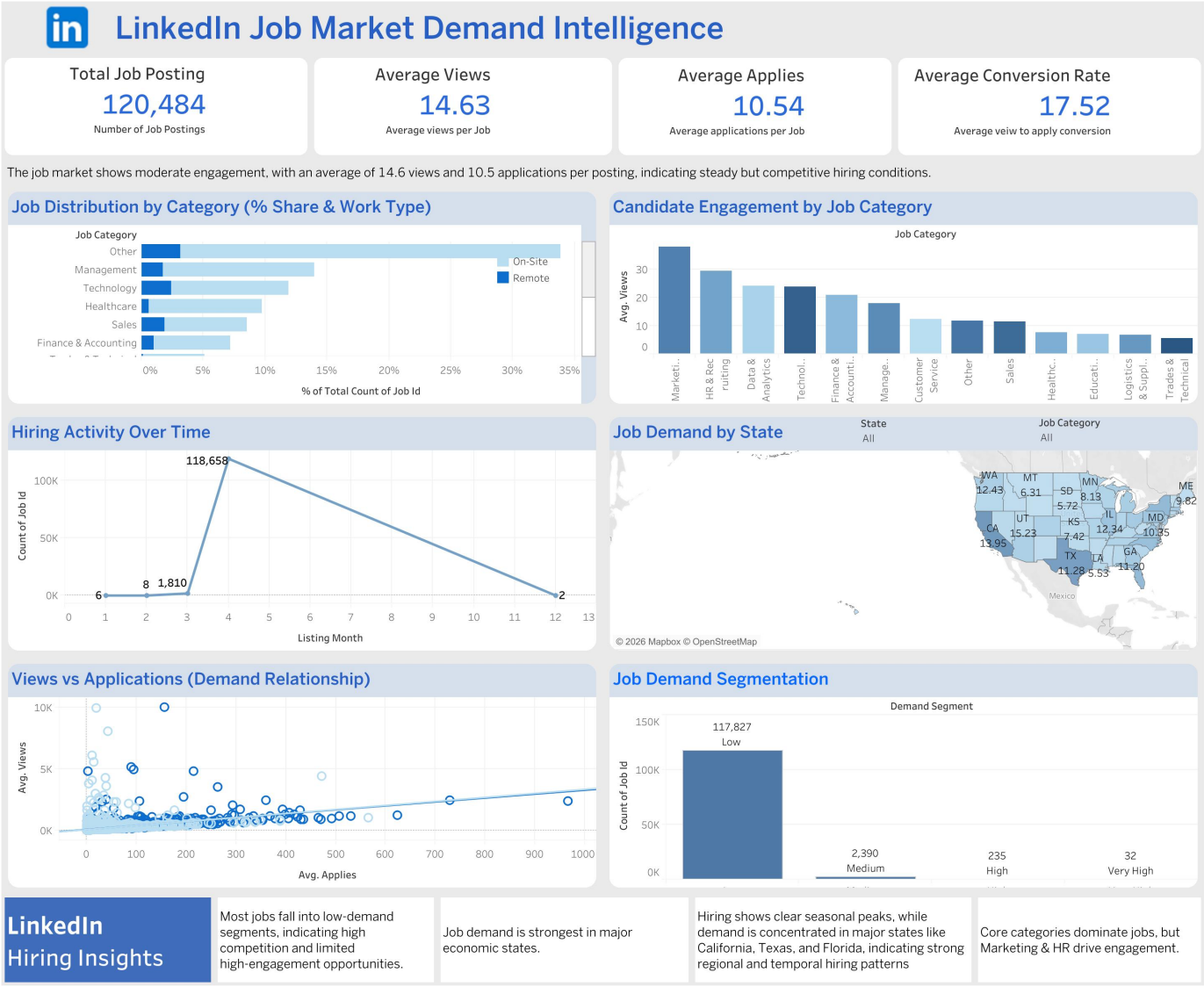

Figure 3. Competition & Opportunity Intelligence - KPI cards and upper analytical view, views with remote competition, salary-demand matrix, and hiring difficulty.

11. Insights Summary

No.	Decision insight
1	The market has scale but moderate engagement: 120,484 postings average only 14.63 views and 10.54 applications, so most roles compete for limited candidate attention.
2	Low-demand postings dominate; employers must improve listing quality, salary disclosure, and targeting rather than rely on volume alone.
3	Marketing and HR & Recruiting attract high engagement, making them strong categories for fast pipeline building but also stronger competition.
4	Data & Analytics and Technology offer the strongest salary opportunities, making them premium skill categories.
5	Salary transparency is weak because only 29.12% disclose usable salary information.
6	Experience level strongly affects pay; Executive and Director roles sit far above Entry-level and Internship roles.
7	Remote jobs show higher salary and competition, expanding opportunity but increasing applicant pressure.
8	California, Texas, and Florida appear as major demand centres.
9	High-view low-application jobs reveal possible problems with role attractiveness or unclear job details.
10	HR & Recruiting and Technology show strong efficiency, converting candidate interest into applications faster.
11	Technology has high competition, requiring stronger screening and candidate differentiation.
12	High-demand jobs are rare; analyze them to learn best practices in role description, salary framing, and application process.

12. Recommendations

Priority	Insight	Recommendation	Expected impact
1	Salary transparency is low.	Increase salary disclosure and salary brackets for priority roles.	Higher trust, better-fit applications, reduced screening waste.
2	Low-demand postings dominate.	Improve job titles, descriptions, skill clarity, and targeting.	Higher conversion and better listing ROI.
3	Data & Analytics and Technology pay more.	Benchmark technical compensation and create premium hiring playbooks.	Improved offer competitiveness.
4	Remote roles have higher competition.	Create separate remote hiring funnels with faster screening and stronger filters.	Better recruiter productivity and reduced backlog.
5	Marketing and HR drive engagement.	Use these categories for rapid hiring campaigns and candidate pool building.	Faster pipeline creation.

13. Impact Estimation

Impact estimates are directional because the analysis uses public posting data. However, at a scale above 120,000 postings, even small improvements in conversion or screening efficiency can create meaningful operational impact.

Action	Approximate impact logic	Estimated value
Improve salary transparency	A small conversion lift across low-demand postings would affect the largest market segment.	Better candidate-job fit and lower screening waste.
Optimize low-demand listings	A 10% improvement in views or applications would improve listing ROI across thousands of postings.	Higher recruiter productivity.
Separate remote hiring funnel	Remote roles show higher competition; filtering separately reduces manual load.	Faster shortlisting.
Target high-opportunity categories	Focus pay and branding on Data & Analytics and Technology.	Higher acceptance probability for skilled roles.

14. Limitations

- The dataset represents LinkedIn postings and may not include jobs posted elsewhere.
- Salary analysis is constrained by missing salary information.
- Views and applications measure engagement, not confirmed hiring.
- Listing-month concentration around April limits seasonality conclusions.
- Broad job categories can hide within-category variation.
- Remote labels may simplify hybrid arrangements.

15. Future Scope

- Add predictive modelling for application probability, hard-to-fill risk, and salary range prediction.
- Build a skill-demand dashboard using the exploded skills dataset.
- Integrate unemployment rate, cost of living, GDP, and industry growth by region.
- Add scheduled refresh for near-real-time dashboards.
- Create a candidate-facing recommendation tool using demand, salary, and competition signals.

16. Conclusion

This project converted a large LinkedIn job posting dataset into a practical market intelligence system for recruitment and career decision-making. Python cleaned and enriched the data, statistical analysis validated key salary and category differences, and Tableau translated the findings into executive-ready dashboards. The strongest conclusion is that the job market is uneven: most postings are low-demand, salary transparency is limited, technical and analytical roles offer stronger pay, and remote roles combine higher opportunity with higher competition. The recommended action is to use segmented hiring intelligence by category, salary, remote status, state, and experience level.

Data dictionary - key fields.

Field	Meaning
job_id	Unique job posting identifier.
company_name	Company associated with the posting.
title	Job title text.
job_category	Grouped role category.
formatted_experience_level	Experience label.
remote_label	Remote or On-Site classification.
state	Derived geographic state.
normalized_salary	Comparable annual salary where available.
salary_bracket	Grouped salary band.
views	Number of listing views.
applies	Number of applications where available.
conversion_rate	Application-to-view conversion indicator.
days_active	Number of active posting days.

Notebook evidence: 02_cleaning.ipynb contains the ETL pipeline; 03_eda_visualizations.ipynb contains visual exploration; 04_statistical_analysis.ipynb contains tests and segmentation; 05_final_load_prep.ipynb contains final Tableau loading and preparation.

18. Contribution Matrix

Fill this section with the real team member names and roles before submission. Claims should match GitHub commits, PRs, notebooks, dashboards, and report evidence.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Palak	✓	✓			✓		
Divya Singh			✓	✓		✓	
Arun		✓					✓
Krishna Pramod Dubale					✓		✓
Samarth Sangtani			✓				✓
Sarvesh Kumar	✓				✓		

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.